

Лабораторная работа №3

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Тема: Настройка виртуальной локальной сети (VLAN)

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

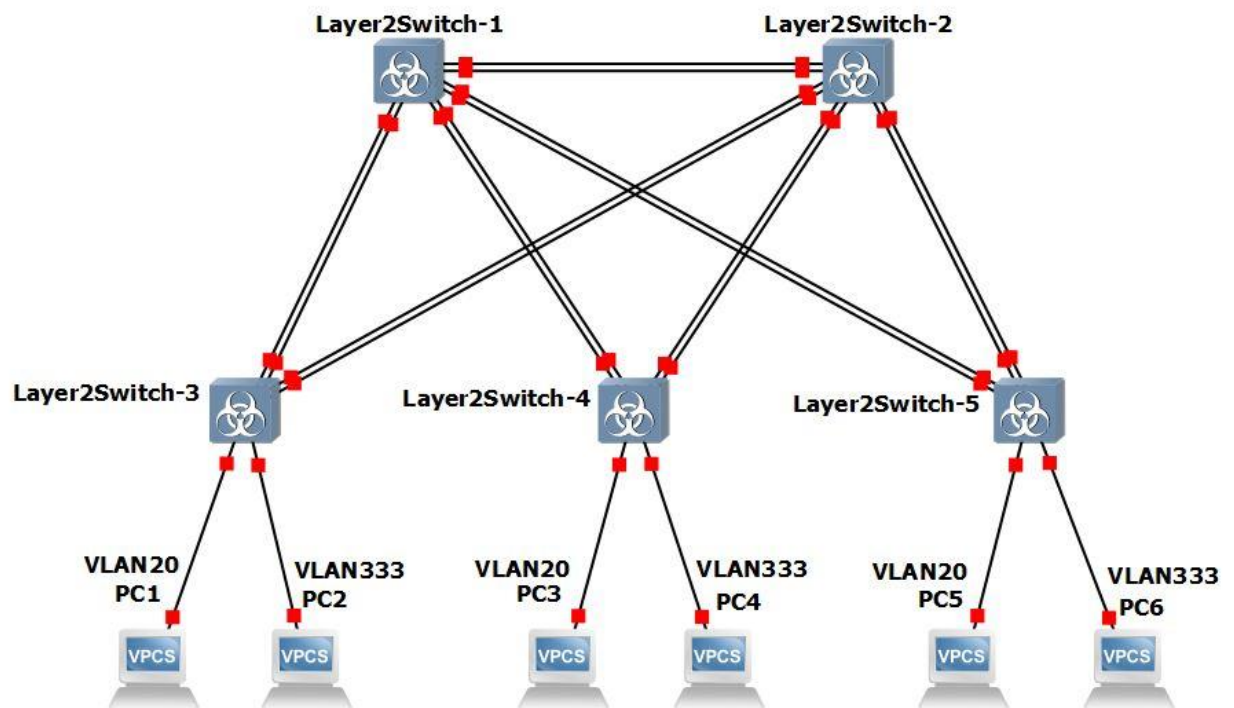
nb! - отметка в тексте, "обратите особое внимание"

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров

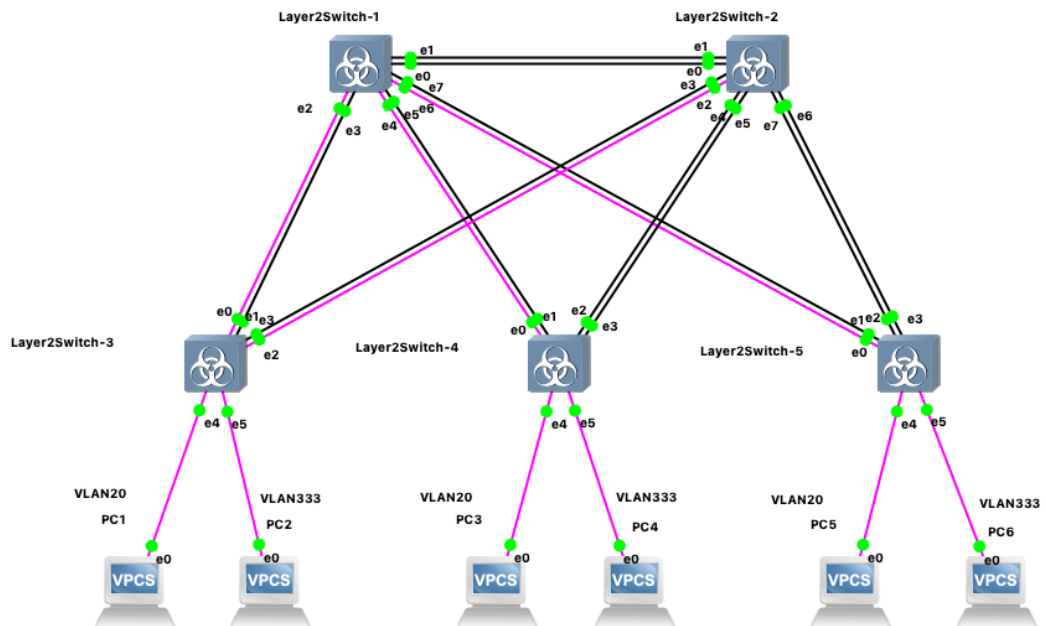
настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между

коммутаторами использовать Native VLAN

Схема сети:



Выделим линки, по которым будут идти данные:



Для каждого PC зададим IP адрес:

PC1> ip 192.168.1.1

PC2> ip 192.168.1.2

PC3> ip 192.168.1.3

PC4> ip 192.168.1.4

PC5> ip 192.168.1.5

PC6> ip 192.168.1.6

Настройка свитчей SW3, SW4, SW5 (везде одинаково:

Switch>enable

Switch#configure terminal

Настройка порта Gi1/0

Switch(config)#interface Gi1/0

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Настройка порта Gi1/1

```
Switch(config)#interface Gi1/1
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 333
```

```
Switch(config-if)#exit
```

Настройка порта Gi0/0 (и Gi0/3 для SW3)

```
Switch(config)#interface Gi0/0
```

```
Switch(config-if)#switchport trunk encapsulation dot1q
```

```
Switch(config-if)#switchport mode trunk
```

```
Switch(config-if)#switchport trunk native vlan 333
```

```
Switch(config-if)#switchport trunk allowed vlan 20,333
```

```
Switch(config-if)#exit
```

Настройка портов Gi0/1, Gi0/2, Gi0/3 (для SW3 Gi0/3 настраивается как Gi0/0)

```
Switch(config)#int range Gi0/1-3
```

```
Switch(config-if-range)#switchport trunk allowed vlan remove 20,333
```

```
Switch(config-if-range)#exit
```

Aug 2 12:07:20.011: %SYS-5-CONFIG-I- Configured from console by console

[Switch#show interface Gi0/3 trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/3	desirable	n-isl	trunking	1

Port	Vlans allowed on trunk
Gi0/3	1-19,21-332,334-4094

Port	Vlans allowed and active in management domain
Gi0/3	1,100,200,300

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/3	100,200,300

[Switch#show interface Gi0/0 trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	333

Port	Vlans allowed on trunk
Gi0/0	20,333

Port	Vlans allowed and active in management domain
Gi0/0	20,333

Port	Vlans in spanning tree forwarding state and not pruned
Gi0/0	20,333

Switch#show interface status

Gi0/0 20,333

[Switch#show interface status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gi1/0		connected	20	auto	auto	unknown
Gi1/1		connected	333	auto	auto	unknown

Switch#show vlan brief

```
[Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
20	VLAN20	active	Gi1/0
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
333	VLAN333	active	Gi1/1
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

Настройка SW1:

```
Switch>enable
```

```
Switch#configure terminal
```

```
Switch(config)#int range gi0/2, gi1/0, gi1/2
```

```
Switch(config-if-range)#switchport trunk encapsulation dot1q
```

```
Switch(config-if-range)#switchport mode trunk
```

```
Switch(config-if-range)#switchport trunk native vlan 333
```

```
Switch(config-if-range)#switchport trunk allowed vlan 20,333
```

```
Switch(config-if-range)#exit
```

```
Switch(config)#int range gi0/0-1, gi1/1, gi1/3
```

```
Switch(config-if-range)#switchport trunk allowed vlan remove 20,333
```

```
Switch(config-if-range)#exit
```

Настройка SW2:

```
Switch>enable
```

```
Switch#configure terminal
```

```
^
```

```
% Invalid input detected at '^' marker.
```

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#int range gi0/0-1, gi0/3, gi1/0-3

Switch(config-if-range)#switchport trunk allowed vlan remove 20,333

Switch(config-if-range)#exit

Switch(config)#interface gi0/2

Switch(config-if)#switchport trunk encapsulation dot1q

Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 333

Switch(config-if)#switchport trunk allowed vlan 20,333

Switch(config-if)#exit

2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать

Для PC1:

```
[PC1> ip 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

[PC1> ping 192.168.1.2

host (192.168.1.2) not reachable

[PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=12.980 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=7.852 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=9.274 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=12.572 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=13.882 ms

[PC1> ping 192.168.1.4

host (192.168.1.4) not reachable

[PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=24.952 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=11.221 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=11.833 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=24.522 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=33.949 ms

[PC1> ping 192.168.1.6

host (192.168.1.6) not reachable
```

Для PC2:

```
[PC2> ping 192.168.1.1

host (192.168.1.1) not reachable

[PC2> ping 192.168.1.3

host (192.168.1.3) not reachable

[PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=16.851 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=13.064 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=15.774 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=11.382 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=9.809 ms

[PC2> ping 192.168.1.5

host (192.168.1.5) not reachable

[PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=12.694 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=24.146 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=23.418 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=13.763 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=9.915 ms
```

Хосты своего VLAN доступны, другого – нет.

3) Перехватить в Wireshark пакеты с тегами и без тегов (nb!), результаты задокументировать

Тегированные пакеты Vlan 20

No.	Time	Source	Destination	Protocol	Length	Info
53	39.062134	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.1.3? Tell 192.168.1.1
54	39.067190	Private_66:68:02	Private_66:68:00	ARP	68	192.168.1.3 is at 00:50:79:66:68:02
55	39.074084	192.168.1.1	192.168.1.3	ICMP	102	Echo (ping) request id=0xbca5, seq=1/256, ttl=64 (reply in 56)
56	39.079073	192.168.1.3	192.168.1.1	ICMP	102	Echo (ping) reply id=0xbca5, seq=1/256, ttl=64 (request in 55)
57	40.080076	192.168.1.1	192.168.1.3	ICMP	102	Echo (ping) request id=0xbda5, seq=2/512, ttl=64 (reply in 58)
58	40.086414	192.168.1.3	192.168.1.1	ICMP	102	Echo (ping) reply id=0xbda5, seq=2/512, ttl=64 (request in 57)
62	41.087529	192.168.1.1	192.168.1.3	ICMP	102	Echo (ping) request id=0xbea5, seq=3/768, ttl=64 (reply in 63)
63	41.093875	192.168.1.3	192.168.1.1	ICMP	102	Echo (ping) reply id=0xbea5, seq=3/768, ttl=64 (request in 62)
64	42.095197	192.168.1.1	192.168.1.3	ICMP	102	Echo (ping) request id=0xbfa5, seq=4/1024, ttl=64 (reply in 65)
65	42.098321	192.168.1.3	192.168.1.1	ICMP	102	Echo (ping) reply id=0xbfa5, seq=4/1024, ttl=64 (request in 64)
68	43.099419	192.168.1.1	192.168.1.3	ICMP	102	Echo (ping) request id=0xc0a5, seq=5/1280, ttl=64 (reply in 69)
69	43.107620	192.168.1.3	192.168.1.1	ICMP	102	Echo (ping) reply id=0xc0a5, seq=5/1280, ttl=64 (request in 68)

> Frame 53: 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface - 1	0000	ff ff ff ff ff ff 00 50 79 66 68 00 81 00 00 14	-----P yfh----
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)	0010	08 06 00 01 08 00 06 04 00 01 00 50 79 66 68 00	-----Pyfh----
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20	0020	c0 a8 01 01 ff ff ff ff ff ff c0 a8 01 03 00 00	-----
> Address Resolution Protocol (request)	0030	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
	0040	00 00 00 00	----

```

✓ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: ARP (0x0806)
    Padding: 00000000000000000000000000000000
    Trailer: 00000000

```

Заголовок VLAN (4 байта):

- 1) Приоритет
- 2) ID vlan
- 3) Тип протокола внутри

Нетегированные пакеты Native Vlan 333:

The image shows a Wireshark packet capture window titled 'arp or icmp'. The packet list on the left shows 14 packets. The packet details pane on the right shows the structure of the selected packet (No. 29, an ARP request). The packet bytes pane at the bottom shows the raw data of the selected packet, with a hex dump and ASCII representation.

No.	Time	Source	Destination	Protocol	Length	Info
29	21.248711	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.4? Tell 192.168.1.2
30	21.257310	Private_66:68:03	Private_66:68:01	ARP	64	192.168.1.4 is at 00:50:79:66:68:03
31	21.257900	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0x59a7, seq=1/256, ttl=64 (reply in 32)
32	21.263554	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0x59a7, seq=1/256, ttl=64 (request in 31)
35	22.265409	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0x5aa7, seq=2/512, ttl=64 (reply in 36)
36	22.271387	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0x5aa7, seq=2/512, ttl=64 (request in 35)
37	23.272844	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0x5ba7, seq=3/768, ttl=64 (reply in 38)
38	23.280204	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0x5ba7, seq=3/768, ttl=64 (request in 37)
42	24.282352	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0x5ca7, seq=4/1024, ttl=64 (reply in 43)
43	24.293902	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0x5ca7, seq=4/1024, ttl=64 (request in 42)
46	25.294271	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0x5da7, seq=5/1280, ttl=64 (reply in 47)
47	25.295686	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0x5da7, seq=5/1280, ttl=64 (request in 46)

Packet Details for Packet 29 (ARP Request):

- Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- Address Resolution Protocol (request)

Packet Bytes for Packet 29:

```

0000  ff ff ff ff ff ff ff 00 50 79 66 68 01 08 06 00 01  ....P yfh....
0010  08 00 06 04 00 01 00 50 79 66 68 01 c0 a8 01 02  ....P yfh....
0020  ff ff ff ff ff ff c0 a8 01 04 00 00 00 00 00 00  ....
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ....
  
```



```
> Frame 29: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, :
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)
```

Нету тега vlan в заголовках.

4) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

5*) Опциональное задание: Добавить в схему маршрутизатор, подключенный к коммутаторам Layer2Switch1 и Layer2Switch2, настроить через него маршрутизацию между VLAN

Полезная информация: избыточные физические каналы можно поместить в отдельные VLAN и обойтись без STP